

HW#2

Homework Report

Course	ECG 762
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Problem 01 3.3

The data $x[n] = Ar^n + w[n]$ for $n = 0, 1, \dots, N-1$ are observed, where $w[n]$ is WGN with variance σ^2 and $r > 0$ is known. Find the CRLB for A . Show that an efficient estimator exists and find its variance. What happens to the variance as $N \rightarrow \infty$ for various values of r ?

Solution:

The observation model is

$$x[n] = Ar^n + w[n], \quad n = 0, 1, \dots, N-1,$$

where $r > 0$ is known and A is the unknown parameter.

Here,

$$s[n; A] = Ar^n, \quad \frac{\partial s[n; A]}{\partial A} = r^n.$$

Substituting into the given formula yields

$$\text{Var}(\hat{A}) \geq \text{CRLB}(A) = \frac{\sigma^2}{\sum_{n=0}^{N-1} r^{2n}}.$$

Using the geometric-series formula,

$$\text{CRLB}(A) = \begin{cases} \sigma^2 \frac{1-r^2}{1-r^{2N}}, & r \neq 1, \\ \frac{\sigma^2}{N}, & r = 1. \end{cases}$$

To show that an efficient estimator exists, consider

$$\hat{A} = \frac{\sum_{n=0}^{N-1} r^n x[n]}{\sum_{n=0}^{N-1} r^{2n}}.$$

Substituting $x[n] = Ar^n + w[n]$ gives

$$\hat{A} = A + \frac{\sum_{n=0}^{N-1} r^n w[n]}{\sum_{n=0}^{N-1} r^{2n}}.$$

Since $\mathbb{E}[w[n]] = 0$,

$$\mathbb{E}[\hat{A}] = A,$$

so the estimator is unbiased.

Using the independence of the noise samples and the identity

$\text{Var}(cX) = c^2 \text{Var}(X)$, we obtain

$$\begin{aligned} \text{Var}(\hat{A}) &= \frac{\sum_{n=0}^{N-1} r^{2n} \text{Var}(w[n])}{(\sum_{n=0}^{N-1} r^{2n})^2} \\ &= \frac{\sigma^2 \sum_{n=0}^{N-1} r^{2n}}{(\sum_{n=0}^{N-1} r^{2n})^2} \\ &= \frac{\sigma^2}{\sum_{n=0}^{N-1} r^{2n}}. \end{aligned}$$

Thus, $\hat{A}$ is unbiased and its variance equals the CRLB. Therefore, it is an efficient estimator. As $N \rightarrow \infty$, the variance behaves as follows:

- If $0 < r < 1$:

$$\text{Var}(\hat{A}) \rightarrow \sigma^2(1 - r^2) > 0.$$

The signal decays with n , so the total Fisher information remains finite. Consequently, the estimation variance does not approach zero.

- If $r = 1$:

$$\text{Var}(\hat{A}) = \frac{\sigma^2}{N} \rightarrow 0.$$

The variance decreases at the rate $1/N$.

- If $r > 1$:

$$\text{Var}(\hat{A}) = \frac{\sigma^2(r^2 - 1)}{r^{2N} - 1} \rightarrow 0.$$

The variance decreases exponentially because later observations contain increasingly strong signals.

Problem 02 3.4

If $x[n] = r^n + w[n]$ for $n = 0, 1, \dots, N - 1$ are observed, where $w[n]$ is WGN with variance σ^2 and r is to be estimated, find the CRLB. Does an efficient estimator exist and if so find its variance?

Solution:

The observation model is

$$x[n] = r^n + w[n], \quad n = 0, 1, \dots, N - 1,$$

where r is the unknown parameter.

Here,

$$s[n; r] = r^n.$$

For $n = 0$, the signal equals 1 and its derivative with respect to r is zero. For $n \geq 1$,

$$\frac{\partial s[n; r]}{\partial r} = n r^{n-1}.$$

Therefore, for $N \geq 2$, the given formula yields

$$\text{Var}(\hat{r}) \geq \text{CRLB}(r) = \frac{\sigma^2}{\sum_{n=1}^{N-1} n^2 r^{2n-2}}.$$

To determine whether an efficient estimator exists, use the equality condition for the CRLB:

$$\frac{\partial \ell(r)}{\partial r} = I(r)(\hat{r} - r),$$

where $\ell(r)$ is the log-likelihood and $I(r)$ is the Fisher information.

For this model,
so

$$\frac{\partial \ell(r)}{\partial r} = \frac{1}{\sigma^2} \sum_{n=1}^{N-1} n r^{n-1} (x[n] - r^n).$$

Let

$$S(r) = \sum_{n=1}^{N-1} n^2 r^{2n-2}, \quad I(r) = \frac{S(r)}{\sigma^2}.$$

The equality condition then requires

$$\hat{r} = r + \frac{\sum_{n=1}^{N-1} n r^{n-1} (x[n] - r^n)}{S(r)}.$$

For $N \geq 3$, the coefficients of $x[1]$ and $x[2]$ in this expression are

$$a_1(r) = \frac{1}{S(r)}, \quad a_2(r) = \frac{2r}{S(r)}.$$

Their ratio is

$$\frac{a_2(r)}{a_1(r)} = 2r.$$

This ratio depends on the unknown parameter. Consequently, no single estimator that depends only on the observations and known quantities can satisfy the equality condition for every r . Thus, **for $N \geq 3$, no unbiased estimator attains the CRLB throughout the parameter range.** There are two special sample-size cases:

- **If $N = 2$:** Since $x[1] = r + w[1]$, the estimator
 $\hat{r} = x[1]$

satisfies

$$\mathbb{E}[\hat{r}] = r, \quad \text{Var}(\hat{r}) = \sigma^2 = \text{CRLB}(r).$$

Hence, an efficient estimator exists and its variance is σ^2 .

- **If $N = 1$:** The only observation is $x[0] = 1 + w[0]$, whose distribution does not depend on r . The Fisher information is zero, and r is not identifiable.

Problem 03 3.9

We observe two samples of a DC level in correlated Gaussian noise

$$\begin{aligned} x[0] &= A + w[0] \\ x[1] &= A + w[1] \end{aligned}$$

where $\mathbf{w} = [w[0]w[1]]^T$ is zero mean with covariance matrix

$$\mathbf{C} = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

The parameter ρ is the correlation coefficient between $w[0]$ and $w[1]$. Compute the CRLB for A and compare it to the case when $w[n]$ is WGN or $\rho = 0$. Also, explain what happens when $\rho \rightarrow \pm 1$. Finally, comment on the additivity property of the Fisher information for nonindependent observations.

Solution:

The two observations can be written as

$$x = A1 + w, \quad 1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

where

$$w \sim \mathcal{N}(0, C), \quad C = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Thus,

$$x \sim \mathcal{N}(\mu, (A)C), \quad \mu(A) = A1.$$

First, assume $-1 < \rho < 1$, so that C is invertible.

Using the given Fisher information formula,

$$I(A) = \left[\frac{\partial \mu(A)}{\partial A} \right]^T C^{-1} \left[\frac{\partial \mu(A)}{\partial A} \right] + \frac{1}{2} \text{tr} \left[\left(C^{-1} \frac{\partial C}{\partial A} \right)^2 \right].$$

Since

$$\frac{\partial \mu(A)}{\partial A} = 1, \quad \frac{\partial C}{\partial A} = 0,$$

the covariance-dependent term vanishes. Therefore,

$$I(A) = 1^T C^{-1} 1.$$

The inverse covariance matrix is

$$C^{-1} = \frac{1}{\sigma^2(1 - \rho^2)} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}.$$

Consequently,

$$\begin{aligned} I(A) &= \frac{1}{\sigma^2(1 - \rho^2)} [1 \quad 1] \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \frac{2(1 - \rho)}{\sigma^2(1 - \rho^2)} \\ &= \frac{2}{\sigma^2(1 + \rho)}. \end{aligned}$$

Hence, the CRLB for any unbiased estimator of A is

$$\text{Var}(\hat{A}) \geq \text{CRLB}(A) = \frac{\sigma^2(1 + \rho)}{2}.$$

This bound is attained by the sample mean

$$\hat{A} = \frac{x[0] + x[1]}{2}.$$

Indeed,

$$\mathbb{E}[\hat{A}] = \frac{\mathbb{E}[x[0]] + \mathbb{E}[x[1]]}{2} = A,$$

so the estimator is unbiased. Its variance is

$$\begin{aligned} \text{Var}(\hat{A}) &= \frac{1}{4} \text{Var}(w[0] + w[1]) \\ &= \frac{1}{4} [\text{Var}(w[0]) + \text{Var}(w[1]) + 2\text{Cov}(w, [0]w[1])] \\ &= \frac{1}{4} (\sigma^2 + \sigma^2 + 2\rho\sigma^2) \\ &= \frac{\sigma^2(1 + \rho)}{2}. \end{aligned}$$

Therefore, $\hat{A}$ is an efficient estimator.

For white Gaussian noise, $\rho = 0$, giving

$$\text{CRLB}(A) = \frac{\sigma^2}{2}.$$

Positive correlation increases the bound relative to the independent-noise case, whereas negative correlation decreases it. Positively correlated noise tends to reinforce itself when averaged; negatively correlated noise tends to cancel.

As $\rho \rightarrow 1$,

$$\text{CRLB}(A) \rightarrow \sigma^2.$$

At $\rho = 1$, the noise samples satisfy $w[1] = w[0]$ almost surely, so
 $x[0] = x[1] = A + w[0]$.

The second observation provides no additional information, and averaging does not reduce the noise variance.

As $\rho \rightarrow -1$,

$$\text{CRLB}(A) \rightarrow 0.$$

At $\rho = -1$, the noise samples satisfy $w[1] = -w[0]$ almost surely. Therefore,

$$\hat{A} = \frac{x[0] + x[1]}{2} = A + \frac{w[0] + w[1]}{2} = A.$$

Problem 04 3.13

Consider a generalization of the line fitting problem as described in Example 3.7, termed polynomial or curve fitting. The data model is $x[n] = \sum_{k=0}^{p-1} A_k n^k + w[n]$

for $n = 0, 1, \dots, N-1$. As before, $w[n]$ is WGN with variance σ^2 . It is desired to estimate $\{A_0, A_1, \dots, A_{p-1}\}$. Find the Fisher information matrix for this problem.

Where 3.7 is: Prove that in Example 3.4

$$\frac{1}{N} \sum_{n=0}^{N-1} \cos(4\pi f_0 n + 2\phi) \approx 0.$$

What conditions on f_0 are required for this to hold? Hint: Note that

$$\sum_{n=0}^{N-1} \cos(\alpha n + \beta) = \operatorname{Re} \left(\sum_{n=0}^{N-1} \exp[j(\alpha n + \beta)] \right)$$

and use the geometric progression sum formula.

Solution:

The observation model is

$$x[n] = \sum_{k=0}^{p-1} A_k n^k + w[n], \quad n = 0, 1, \dots, N-1,$$

where

$$w[n] \sim \mathcal{N}(0, \sigma^2).$$

The unknown parameter vector is

$$A = [A_0 \quad A_1 \quad \dots \quad A_{p-1}]^T.$$

Each observation has the distribution

$$x[n] \sim \mathcal{N}(\sum_{k=0}^{p-1} A_k n^k, \sigma^2).$$

Since the observations are independent, their joint likelihood is

$$p(x; A) = (2\pi\sigma^2)^{-N/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} \left(x[n] - \sum_{k=0}^{p-1} A_k n^k \right)^2 \right].$$

Therefore, the log-likelihood is

$$\ell(A) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} \left(x[n] - \sum_{k=0}^{p-1} A_k n^k \right)^2.$$

For $i = 0, 1, \dots, p-1$,

$$\frac{\partial}{\partial A_i} \left(\sum_{k=0}^{p-1} A_k n^k \right) = n^i.$$

Thus,

$$\frac{\partial \ell}{\partial A_i} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} n^i \left(x[n] - \sum_{k=0}^{p-1} A_k n^k \right).$$

Differentiating again with respect to A_j gives

$$\frac{\partial^2 \ell}{\partial A_i \partial A_j} = -\frac{1}{\sigma^2} \sum_{n=0}^{N-1} n^{i+j}.$$

The (i, j) entry of the Fisher information matrix is

$$\mathcal{I}_{ij}(A) = -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial A_i \partial A_j} \right].$$

Since the second derivative does not depend on the observations,

$$\mathcal{I}_{ij}(A) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} n^{i+j}, \quad i, j = 0, 1, \dots, p-1.$$

Define

$$S_m = \sum_{n=0}^{N-1} n^m, \quad S_0 = N,$$

where the constant basis function n^0 is taken to be 1, including at $n = 0$. The Fisher information matrix is therefore

$$\mathcal{I}(A) = \frac{1}{\sigma^2} \begin{bmatrix} S_0 & S_1 & \cdots & S_{p-1} \\ S_1 & S_2 & \cdots & S_p \\ \vdots & \vdots & \ddots & \vdots \\ S_{p-1} & S_p & \cdots & S_{2p-2} \end{bmatrix}.$$

Equivalently, defining the design matrix

$$H = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & \cdots & 2^{p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & N-1 & (N-1)^2 & \cdots & (N-1)^{p-1} \end{bmatrix},$$

the result can be written as

$$\mathcal{I}(A) = \frac{1}{\sigma^2} H^T H.$$

For example, when $p = 2$, the model reduces to straight-line fitting:

$$x[n] = A_0 + A_1 n + w[n].$$

In this case,

$$\mathcal{I}(A) = \frac{1}{\sigma^2} \begin{bmatrix} N & \frac{N(N-1)}{2} \\ \frac{N(N-1)}{2} & \frac{N(N-1)(2N-1)}{6} \end{bmatrix}.$$

Because the sampling locations $0, 1, \dots, N-1$ are distinct, H has full column rank when $N \geq p$, and the Fisher information matrix is invertible. When $N < p$, the matrix is singular, and the full coefficient vector is not identifiable.

Problem 05 3.14

For the data model in Example 3.11 consider the estimator $\hat{\sigma}_A^2 = (\hat{A})^2$, where $\hat{A}$ is the sample mean. Assume we observe a given data set in which the realization of the random variable A is the value A_0 . Show that $\hat{A} \rightarrow A_0$ as $N \rightarrow \infty$ by verifying that

$$E(\hat{A} \mid A = A_0) = A_0$$

$$\text{var}(\hat{A} \mid A = A_0) = \frac{\sigma^2}{N}.$$

Hence, $\hat{\sigma}_A^2 \rightarrow A_0^2$ as $N \rightarrow \infty$ for the given realization $A = A_0$. Next find the variance of $\hat{\sigma}_A^2$ as $N \rightarrow \infty$ by determining $\text{var}(A^2)$, where $A \sim \mathcal{N}(0, \sigma_A^2)$, and compare it to the CRLB. Explain why σ_A^2 cannot be estimated without error even for $N \rightarrow \infty$.

Solution:

Consider the random-amplitude model

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1,$$

where

$$A \sim \mathcal{N}(0, \sigma_A^2), \quad w[n] \sim \mathcal{N}(0, \sigma^2).$$

Assume that A is independent of the noise and that σ^2 is known. All N observations share the same realization of A .

The sample mean is

$$\hat{A} = \frac{1}{N} \sum_{n=0}^{N-1} x[n],$$

and the proposed estimator of σ_A^2 is

$$\widehat{\sigma}_A^2 = (\hat{A})^2.$$

Conditional on the realization $A = A_0$,

$$x[n] = A_0 + w[n],$$

so

$$\hat{A} = A_0 + \frac{1}{N} \sum_{n=0}^{N-1} w[n].$$

Since the noise is independent of A and has zero mean,

$$\begin{aligned} \mathbb{E}[\hat{A} \mid A = A_0] &= A_0 + \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}[w[n] \mid A = A_0] \\ &= A_0. \end{aligned}$$

Furthermore, independence of the noise samples gives

$$\begin{aligned} \text{Var}(\hat{A} \mid A = A_0) &= \text{Var}\left(\frac{1}{N} \sum_{n=0}^{N-1} w[n]\right) \\ &= \frac{1}{N^2} \sum_{n=0}^{N-1} \text{Var}(w[n]) \\ &= \frac{\sigma^2}{N}. \end{aligned}$$

Consequently,

$$\mathbb{E} \left[(\hat{A} - A_0)^2 \mid A = A_0 \right] = \frac{\sigma^2}{N} \rightarrow 0.$$

Thus, conditional on $A = A_0$, the estimator $\hat{A}$ converges to A_0 in mean square and hence in probability:

$$\hat{A} \xrightarrow{P} A_0.$$

By continuity of the squaring function,

$$\widehat{\sigma_A^2} = (\hat{A})^2 \xrightarrow{P} A_0^2.$$

The limit is the squared amplitude of the particular realization, rather than necessarily the population variance σ_A^2 .

To determine the variation across realizations, note that a zero-mean Gaussian random variable satisfies

$$\mathbb{E}[A^2] = \sigma_A^2, \quad \mathbb{E}[A^4] = 3\sigma_A^4.$$

Therefore,

$$\begin{aligned} \text{Var}(A^2) &= \mathbb{E}[A^4] - (\mathbb{E}[A^2])^2 \\ &= 3\sigma_A^4 - \sigma_A^4 \\ &= 2\sigma_A^4. \end{aligned}$$

The finite-sample variance confirms this limit. Since

$$\hat{A} = A + \frac{1}{N} \sum_{n=0}^{N-1} w[n]$$

is the sum of independent Gaussian random variables,

$$\hat{A} \sim \mathcal{N} \left(0, \sigma_A^2 + \frac{\sigma^2}{N} \right).$$

Hence,

$$\text{Var}(\widehat{\sigma_A^2}) = 2 \left(\sigma_A^2 + \frac{\sigma^2}{N} \right)^2,$$

and

$$\lim_{N \rightarrow \infty} \text{Var}(\widehat{\sigma_A^2}) = 2\sigma_A^4.$$

To compare this result with the CRLB, let

$$\tau = \sigma_A^2 > 0.$$

After marginalizing over A , the observation vector has distribution

$$x \sim \mathcal{N}(0, C(\tau)),$$

where

$$C(\tau) = \tau 11^T + \sigma^2 I_N.$$

Here, $\mathbf{1}$ is the N -dimensional vector of ones. The observations are correlated because they share the same random amplitude.

Since the mean does not depend on τ , the Gaussian Fisher information formula reduces to

$$I(\tau) = \frac{1}{2} \text{tr} \left[\left(C^{-1} \frac{\partial C}{\partial \tau} \right)^2 \right].$$

We have

$$\frac{\partial C}{\partial \tau} = \mathbf{1} \mathbf{1}^T,$$

and

$$C \mathbf{1} = (\sigma^2 + N\tau) \mathbf{1},$$

which implies

$$C^{-1} \mathbf{1} = \frac{1}{\sigma^2 + N\tau}.$$

Therefore,

$$\begin{aligned} I(\tau) &= \frac{1}{2} (\mathbf{1}^T C^{-1} \mathbf{1})^2 \\ &= \frac{1}{2} \left(\frac{N}{\sigma^2 + N\tau} \right)^2 \\ &= \frac{1}{2(\tau + \sigma^2/N)^2}. \end{aligned}$$

The CRLB for unbiased estimation of σ_A^2 is consequently

$$\text{CRLB}(\sigma_A^2) = 2 \left(\sigma_A^2 + \frac{\sigma^2}{N} \right)^2.$$

As $N \rightarrow \infty$,

$$\text{CRLB}(\sigma_A^2) \rightarrow 2\sigma_A^4.$$

Thus, the limiting variance of the proposed estimator equals the limiting CRLB.

For finite N , however,

$$\mathbb{E}[\widehat{\sigma_A^2}] = \sigma_A^2 + \frac{\sigma^2}{N},$$

so the estimator is biased. An unbiased version is

$$\widetilde{\sigma_A^2} = (\hat{A})^2 - \frac{\sigma^2}{N},$$

which has the same variance and attains the finite-sample CRLB.